

Network Intrusion Detection using Deep Reinforcement Learning

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Abstract—The number of internet-connected systems has increased by a huge amount in recent years. These systems are highly vulnerable and increasingly at risk from cyber-attacks. These cyber-attacks need more advanced and intelligent cyber defense systems that include agents that can learn to make judgements on their own without consulting human specialists. Deep learning-based reinforcement learning (DRL) is very good at handling complicated, dynamic, and especially high-dimensional cyber protection problems. This article provides a modern network intrusion detection technology, which employs a deep feed-forward neural network method and reinforcement learning, which is based on Q-learning. In order to detect various sorts of intrusions in the network using an automated trial- and-error method and continually improve its detection skills, the Deep Q-Learning (DQL) model is proposed. The accuracy of the model proposed is 91.4%, while the accuracy of other self-taught learning models is 88.4% and it is a similar case for recall rate and precision as well which are 90.2% and 92.8%. Our experimental findings further demonstrate that our suggested DQL beats other comparable machine learning methods and is very good at identifying various intrusion classifications.

Index Terms – Deep learning, neural networks, Network intrusion detection, NSL-KDD, reinforcement learning, deep q-learning, Epsilon Greedy, NSL-KDD, Artificial Intelligence

I. INTRODUCTION

In order to tackle threats brought on by an expanding number of sophisticated hackers in a digital environment, the next generation of intrusion detection systems (IDSs) demand network intrusion detection methods that are intelligent as well as automated [2], and [3]. Specifically, there has been a high demand for automated IDS strategies that are agent-based and need as little involvement of humans as possible while still having the ability to improve and evolve (for example, by taking the right actions at the right time in the environment given) and to be more resilient and robust to unforeseen and potential threats or difficulties that were not seen by it before.

Using automated agents, Reinforcement Learning (RL) has gained popularity as a method for identifying and categorizing various attacks and cyber threats [1]. Two fundamental elements make up a traditional DRL framework: the environment and an agent. The agent has the ability to observe the condition of the environment and take action. The environment responds to the activity by creating a reward and a new state that are related to the outcome of the action, whether it was good or bad.

The following is a summary of our work's significant contributions: The modern methods are presented for detecting network intrusions that integrate deep feed-forward neural networks and reinforcement learning which is based on Q-learning. Our suggested model has the ability to continuously learn how to interact in a network environment so that it can identify various kinds of network invasions. Our model's self-learning capabilities enable it to continuously improve its detecting abilities.

To better network intrusion detection activities, improved self-learning and engagement with the distributed environment are required, intrinsic details of the top methods are presented for adjusting various hyper parameters for reinforcement learning techniques based on deep learning.

The results of the experiment which uses the NSL-KDD datasets show that the model (based on DQL) proposed is quite efficient in classifying the network intrusion attacks and performs better than other comparable machine learning and deep learning approaches, reaching greater than 91% accuracy in the tasks related to classification involved in many network intrusion classes.

II. LITERATURE SURVEY

This section reviews network intrusion detection using deep reinforcement learning. Numerous researchers have worked on it. According to Guillermo Caminero and Manuel Lopez network, Intrusion Detection is an important aspect of today's Network traffic, and it is very important to design proper methods, algorithms, and techniques for proper detection of intrusion in the network [1]. Frameworks for intrusion detection rely on a supervised learning paradigm and a training data set made up of network characteristics and their corresponding intrusion labels. In this study, we combine this paradigm with a reinforcement learning method that typically relies on interacting with a real-world setting. RL algorithms show poor performance when the data set is not properly balanced. This dilemma is the fundamental motivation for proposing the DQN algorithm.

It was suggested by Ying-Feng Hsu and Morito Matsuoka that the total worldwide IP traffic was found to be 120 exabytes per month in the year 2012 and this number is expected to increase to 400 exabytes per month in the year 2023.

[2]. Due to this huge increase in network traffic data, it is essential to provide tools and techniques to avoid network-based attacks. This paper proposed an ANIDS (Anomaly Network Intrusion Detection Model) model for detecting network intrusion attacks. The design of this DRL model is based on a deep Q network (DQN agent) which used a deep neural network for the calculation of the expected value of rewards.

In this study, Hooman Alavizadeh, Julian Jang-Jaccard, and Hootan Alavizadeh employed a feed-forward neural network. The authors suggested a model that combines a deep feed-forward neural net with Q-learning-based relevance.

feedback for network intrusion detection [3]. With the use of an automatic trial-error method, our suggested DQL model can identify different network intrusions, and it continuously improves its detection skills.

Manuel Lopez-Martin, Belen Carro and Antonio Sanchez-Esguevillas proposed in this paper for network intrusion detection are developed using DLR algorithms [4]. They have developed a unique method that conceptually modifies the traditional DRL paradigm (based upon engagement with a live situation), with the environment replaced by a sample function of stored training incursions. Along with sampling the training dataset, this new fake environment also produces incentives for detection mistakes discovered during training. The DLR models used here are Policy Gradient (PG) Deep Q-network (DQN), Double Deep Q-Network (DDQN).

III. PROPOSED SYSTEM

It provides a new model called DQN-RL that puts the RL theory-based classifier into practice. An environment in an RL framework notifies an agent regarding the environment's state, and the response is given back by the agent by taking an action within the environment [4] and [5]. The environment gradually changes because of these actions. The environment reacts to the action done by the agent by giving the agent a reward based on how effective the action was in achieving a certain goal.

By making the a fore mentioned adjustments, a new framework was created that enables the use of a popular RL algorithm (DQN algorithm) to categorize intrusions using a pool of previously collected intrusion data. DQN algorithm is based on determining the agent's (in our case it's the classifier) ideal Q- function [1]. Each state-action combination is assigned a value by a Q-function; this value is the total of the rewards for the state under consideration if we execute the provided action and continue with the present course of action. The fact that the Q- function may be determined repeatedly using the following equation is a significant result:

$$Q(S_i, A_i) \leftarrow Q(S_i, A_i) + \alpha [R_{i+1} + \gamma \max_A Q(S_{i+1}, A_{i+1}) - Q(S_i, A_i)]$$

S_i is the present state, A_i is the current action, S_{i+1} is the future action, R_{i+1} is the future reward value, α is the rate of learning and γ is the discount factor. As in this case the states are not correlated with each other, the discount factor is set.

close to zero and since there is no correlation among the states there is no need for the algorithm to keep track of previous states. and values have been set to 0.1 and 0.001 respectively. Q-function is approximated using a fully connected neural network. The present state, which is related to the characteristics extrapolated from the labeled datasets, serves as the input to this NN. The Q function for the possible actions is represented by the NN's output. The Q-function relates to how effective a particular action would be in the present situation. The aforesaid premises lead to a framework that is based on the DQN algorithm and offers effective results in detecting the attacks.

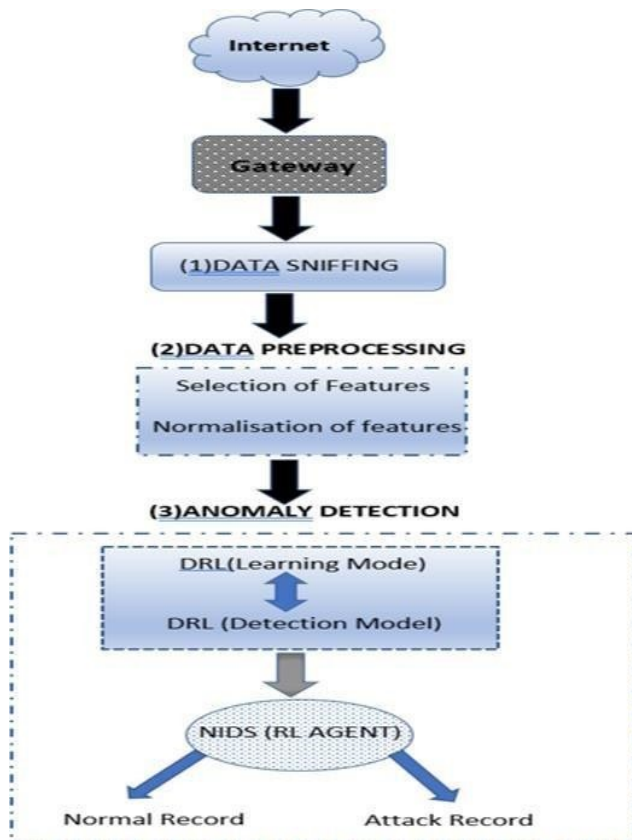


Fig-1: Architecture of Proposed System.

Algorithm

Algorithm for NIDS using Deep reinforcement learning which uses DQN algorithm.

Input: NSL-KDD data set

Output: Attack Detection, Accuracy

Step 1: Loading and pre-processing the data.

Feature Selection: Kendall's rank co-efficient (non-linear) feature selection model is being used for feature selection.

Feature Normalization:

Normalize the data using the min-max normalization.

Data can be normalized by using any one of the following ways-

$$S_{\max} = \frac{\log(s+1)}{\log(s+1)_{\max}}$$

where s is the original raw value.

For the features related to low standard deviation, we normalize them by max values shown below-
 $S_{\text{new}} = \frac{S}{S_{\max}}$

$S_{\max}$

Step 2: Modelling the data using the DQN algorithm.

- A random value is used as the beginning value for the environment's & classifier's Q
- function and is given as $Q_e(s, a_{et})$ and $Q_e(s, a_{et})$. To feed the Q function and derive this state's action values, an initial s_0 state is furthermore picked at random from among all the states in the data set.
- The environment then selects an action a_{et} (intrusion label) in accordance with its policy and the current state.
- Repeat (for every episode)
- Set the state to s_0 , which stands for **random sampling** (data set)-Using a policy derived from $Q_e(s_t, a_{et})$, select the environment's initial action-Perform random sampling ($S(a_{et})$), (where $S(a_{et})$ are all the samples with the label set) and replace the value of S_t with this value.

Repeat (for each step-in $t \rightarrow 0$ to N):

- 1) Set the agent's action act by the policy which is derived from $Q_e(S_t, act)$.
- 2) Now obtain (r_{ct}, ret, S_{t+1}) by taking an RL step: Get the value of r_{ct}, ret which corresponds rewards from the agent and environment. From the policy derive from $Q_e(S_t, a_{et})$ choose the next environment's action which is $act+1$. Perform random Sampling of $(S(a_{et}+1))$, (where $S(a_{et}+1)$ are all the samples which have the label $ae+1$) & replace the value of S_{t+1} with this value.

3) Updating the Q-function:

Calculate the loss function, it is

given by $(\|c_t\| + \max_{c_t \in \{0,1\}} \|c_t\| + 1 - \max_{c_t \in \{0,1\}} \|c_t\|)^2$ and apply gradient descent on it.
 $(\|c_t\| + \max_{c_t \in \{0,1\}} \|c_t\| + 1 - \max_{c_t \in \{0,1\}} \|c_t\|)^2$ and apply gradient descent on it.

Step 3: Classifying the record.

If the record is normal, then the class label is Normal.

Else classifying into the specific class label which may fall under R2L, Probe, Dos.

IV. RESULT

The graph below shows that our suggested DRL model beats other baseline techniques in classifying attacks with an accuracy of 92% and a recall of 91%. The three traditional models in our study—RF, SVM, and MLP [6]—behave similarly in terms of categorization. Because they all make a lot of false-negative predictions, this may be the reason for their collectively low recall rates. The false negative signifies misclassifying regular traffic as aberrant traffic. This reduces system dependability and increases the danger of network intrusion. The experimental finding demonstrates that our method balances the assessment criteria of accuracy, recall, and precision by roughly 92%. We can draw the conclusion that our proposed solution is promising in this use case based on the comparison mentioned previously.

ACCURACY

Accuracy is a factor to consider when ranking classification.

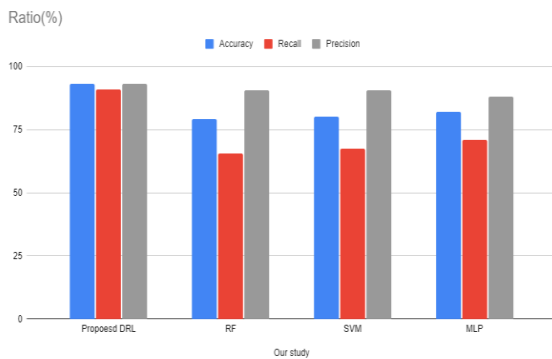


Fig-2: - Accuracy, Recall and Precision

Accuracy is the percentage of forecast that our model correctly anticipated. The following is the actual definition of

Accuracy: The total number of correct forecasts equals the total number of correct predictions.

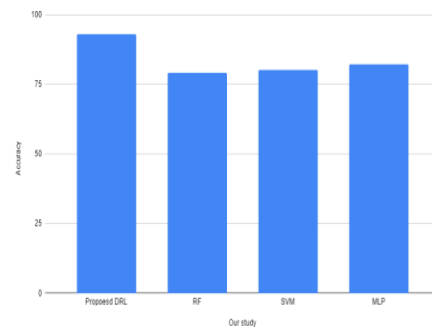


Fig 3: - Sample Accuracy

RECALL

The recall is calculated by dividing the portion of Samples tested that were properly recognized as Positive by the total number of Positive samples. The recall metric measures how successfully the model can detect positive samples. The greater the number of positive samples detected, the greater the recall.

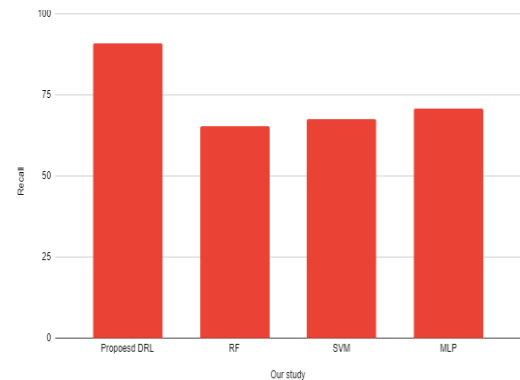


Fig 4: - Sample Recall

PRECISION

Precision, or the caliber of a correct forecast the model makes, is one measure of the model's performance. To calculate precision, divide the overall positive predictions by the proportion of true positives.

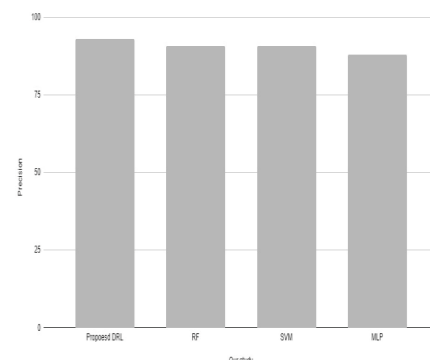


Fig 5: - Sample Precision

V. CONCLUSION

IDS is a crucial component of contemporary data networks. With a model that applies the RL framework to the IDS problem, this work aims to offer a fresh option. The RL algorithms have demonstrated exceptional performance in a variety of fields (such as robotics, banking, gaming, and business operations), and we can demonstrate that they may also be applicable to IDS. The suggested novel model (DQN) combines reinforcement learning with Q learning to create a simulated environment that adheres to RL environment rules. The contributions of this paper are: Introducing an innovative architecture that is based on a fusion of adversarial RL and DQL models, making it an ideal replacement for prediction issues in extremely demanding networks (like IoT networks). Proving the new model's prediction performance is comparable to that of highly non-linear models while boasting the amazing benefit of requiring substantially less computing time each prediction. With an emphasis on prediction performance, prediction timeframes, and model flexibility, a detailed comparison of proposed model with a number of Machine Learning (ML) models are provided. Introducing a model that is especially appropriate for extremely imbalanced datasets since the samples associated with wrongly categorized labels are more frequently used in the training process.

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